# Alex Lupsasca

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Department of Physics & Astronomy
and Department of Mathematics
Vanderbilt University

## Research Interests

Black hole imaging, electromagnetic observations, and (very-long-baseline) radio interferometry

General relativity, gravitational lensing, photon rings, and black hole ringdown

Relativistic astrophysics of compact objects: black holes, neutron stars, and pulsars

Artificial Intelligence: models for reasoning and automated scientific research

## Professional Experience

Research Scientist at OpenAI (2025–)

Project Scientist for the Black Hole Explorer (BHEX) space mission (2024–)

Assistant Professor of Physics & Mathematics, Vanderbilt University (2022–)

Associate Research Scholar, Princeton Gravity Initiative (2020–2022)

Junior Fellow, Harvard Society of Fellows (2017–2020)

## Education

Ph.D., Physics, Harvard University [advisor: Andrew Strominger] (2017)
Thesis: The Maximally Rotating Black Hole as a Critical Point in Astronomy

A.M., Physics, Harvard University (2012)

A.B., Physics & Mathematics, Harvard College (2011)

## Government Funding

U.S.-Israel Binational Science Foundation (**BSF**)
Start-up award of $180,000 (grant number 2024239) to be shared with Israeli co-PI (2025–2027)
Currently supporting collaboration with co-PI Shahar Hadar from the University of Haifa

U.S. National Science Foundation (**NSF**), Division of Physics (PHY)
**CAREER** award of $414,762 (grant number 2340457) in Gravity Theory (2023–2028)
Currently supporting Vanderbilt University postdoctoral researcher Roman Berens

U.S. National Science Foundation (**NSF**), Division of Astronomical Sciences (AST)
Award of $287,554 (grant number 2307888) in Stellar Astronomy & Astrophysics (2023–2026)
Currently supporting Vanderbilt University graduate student Trevor Gravely

## Private Funding

**Simons Foundation**, Simons Collaboration on Black Holes and Strong Gravity (one of 12 co-PIs)
Jointly awarded $8M over 4 years (renewable for 3 more), with $548,414 for my group (2025–2029)

Initial funding for development of BHEX from Fred Ehrsam
Raised $350,000 in seed funding for BHEX at Vanderbilt University (2024)

Research funding from Will and Kacie Snellings
Raised $300,000 in private funding to hire a postdoctoral researcher at Princeton Unversity in 2021
Used to support Alejandro Cárdenas-Avendaño (now faculty at Wake Forest University)

Research funding from the Jacob Goldfield Foundation
Raised $300,000 in private funding to hire a postdoctoral researcher at Harvard University in 2018
Used to support Shahar Hadar (now faculty at the University of Haifa, Israel)

## Prizes & Fellowships

Jon C. Graff, Ph.D. Prize for Excellence in Science Communication from the Society for Science, 2025

Cornelius Vanderbilt Dean's Faculty Fellow in Physics and Astronomy, 2025–2027

**Young Scientist Prize in General Relativity and Gravitation** from the International Union of Pure and Applied Physics and International Society on General Relativity and Gravitation, 2024

**New Horizons in Physics Prize** from the Breakthrough Prize Foundation, 2024

**Junior Fellowship** from the Harvard Society of Fellows, 2017–2020

Joshua S. Friedman Graduate Student Dissertation Fellowship (Harvard University), 2015–2016

Donald and Cathleen Pfister Prize (Harvard College), 2011
Awarded to a graduating senior for excellent achievement in the natural sciences

## Teaching & Advising

Courses taught as faculty at Vanderbilt University:
Physics 1020 (undergraduate): Physics For Future Leaders, Fall 2024
Physics 8021 (graduate): Advanced Electrodynamics, Fall 2023
Physics 8160 (graduate): General Theory of Relativity, Spring 2023 & Spring 2024
Physics 3600 (undergraduate): Seminar in Presenting Physics Research, Fall 2022

Resident Tutor in Kirkland House (Harvard University), 2012-2018
Academic advisor for 5-7 sophomores every year

Star Family Prize Nominee for Excellence in Advising, 2013 & 2015
Derek Bok Center Certificate of Distinction in Teaching, Fall 2012 & Fall 2013

## Mentorship & Supervision

**Postdoctoral researchers**

Roman Berens, Vanderbilt University (2023–)

Alejandro Cárdenas-Avendaño, Princeton University (2021–2023)
Now faculty at Wake Forest University

Shahar Hadar, Harvard University (2018–2020)
Now faculty at the University of Haifa

**Graduate students**

Mirko Landivar Gareca, Vanderbilt University (2025–)

Trevor Gravely, Vanderbilt University (2022–)

Co-supervised at the University of California, Santa Barbara: Joseph Farah (2024–)

Co-supervised at Princeton University: He Jia (2023–), Hengrui Zhu (2021–2022)

Co-supervised at the Observatoire de Paris: Hadrien Paugnat (2021–2022, now at UCLA)

Co-supervised at Harvard University: Delilah Gates (2018–2021, now at Harvard BHI), Mina Himwich (2019–2020, now at Princeton PCTS) and Yichen Shi (2017–2019)

**Undergraduate students**

Lennox Keeble, Princeton University (2024–)

## Service

Referee for:
*Classical and Quantum Gravity* (CQG), *Journal of Cosmology and Astroparticle Physics* (JCAP), *Journal of High Energy Physics* (JHEP), *Physics Letters B* (PLB), *Physical Review D* (PRD), *Physical Review Letters* (PRL), *Physical Review X* (PRX), *The Astrophysical Journal* (ApJ), *European Physical Journal C* (EPJC), *General Relativity and Gravitation* (GRG)

Organizer for:
The Fall 2025 BHEX Science Workshop (online), September 2025
The Photon Ring Science Workshop at Vanderbilt University, February 2024

## Selected Software

`BlackHoleVision` (led by T. Gravely): an interactive iOS app that simulates black hole lensing
`Metric-Reconstruction` (led by R. Berens): linearized metric perturbations of a Kerr black hole
`AART` (led by A. Cárdenas-Avendaño): an Adaptive Analytical Ray Tracing code for accretion disks

## Invited Talks

Royal Netherlands Academy of Arts and Sciences, *Dutch BH Consortium meeting*, July 2026
Institute for Advanced Study, *Workshop on Black Holes from Theory to Observations*, May 2026
Simons Foundation, *Simons Collaboration on Black Holes and Strong Gravity*, May 2026

Simons Foundation, **Simons Foundation Presidential Lecture**, March 2026
IPAM & UCLA, Panel on *Accelerating Math and Theoretical Physics with AI*, March 2026
Harvard University, **Physics Department Special Colloquium**, February 2026
American Association for the Advancement of Science, Annual Meeting Invited Talk, February 2026
OpenAI, Company-wide Talk, September 2025
CERN, **Theory Colloquium**, September 2025
Université de Genève, Department of Physics, *Lunch Bite Seminar*, September 2025
Institut de Radioastronomie Millimétrique, *"BHEX en France" Workshop*, September 2025
Niels Bohr Institute, *Current Themes in Particle Physics and Astrophysics*, August 2025
Harvard Black Hole Initiative, **2025 Annual Conference**, May 2025
Boston Museum of Science, *Peter Galison's 70th Orbit: Celebrating Collaborations*, May 2025
University of Memphis, **Physics Department Colloquium**, April 2025
Princeton Center for Theoretical Science, *Black Hole Mimickers Workshop*, March 2025
New York University, *Astrophysics and Relativity Seminar*, February 2025
Los Alamos National Laboratory, **Astrophysics Distinguished Seminar**, February 2025
University of California, Berkeley, **Astronomy Department Colloquium**, January 2025
Las Cumbres Observatory, *Astronomy Seminar*, January 2025
Erwin Schrödinger Institute, *Lensing and Wave Optics in Strong Gravity Workshop*, December 2024
New York University, *Astrophysics and Relativity Seminar*, November 2024
University of Illinois Urbana-Champaign, *Astrophysics, Relativity, & Cosmology Seminar*, October 2024
Troy University, *Troy University Center for Relativity and Cosmology Seminar*, September 2024
Vanderbilt University, **Mathematics Department Colloquium**, August 2024
Joint Israeli Seminar Series on Gravitational Physics, July 2024
National Astronomical Observatory of Japan, *Black Hole Explorer Japan Workshop*, June 2024
Simons Collaboration on Extreme Electrodynamics of Compact Sources, *BH seminar*, March 2024
Southern Denmark University, *CP3-Origins Seminar*, March 2024
Johns Hopkins University, **Physics Department Colloquium**, January 2024
American Physical Society, Southeastern Section, *Invited Talk*, November 2023
NASA Goddard, *Infrared Science and Technology Integration Group Webinar*, November 2023
Astronomy on Tap, Nashville, October 2023
University of California, Santa Cruz, *Astronomy Group Talk*, October 2023
University of California, Santa Cruz, **Physics Department Colloquium**, October 2023
Institut de Physique Théorique, CEA Saclay, *Conference on Black-Hole Microstructure V*, June 2023
Harvard Black Hole Initiative, *Instrument Design Lab for Space VLBI*, May 2023
Institute for Advanced Study, *IAS Astrophysics Coffee Talk*, May 2023
CERN, *Theory Seminar*, March 2023
Dartmouth College, **Physics Department Colloquium**, February 2023
Carnegie Mellon University, **Physics Department Colloquium**, November 2022
Stanford Institute for Theoretical Physics, *It from Qubit Seminar*, October 2022
Stanford University, **Physics Department Colloquium**, October 2022
Harvard Black Hole Initiative & CMSA, *Conference on Flat Holography*, June 2022
McGill University, *High-Energy Theory Group Meeting*, May 2022
Harvard Black Hole Initiative, **2022 Annual Conference**, May 2022
Harvard University, *High-Energy Physics Seminar*, March 2022
University of Amsterdam, *String Theory Seminar*, March 2022
University of Cambridge, *DAMTP GR Seminar*, March 2022
University of Mississippi, **Physics Department Colloquium**, November 2021
Aveiro University, *Gravity and Gravitational Dynamics Seminar*, October 2021
Albert Einstein Institute (MPI Potsdam), *Astrophysics and Relativity Seminar*, October 2021
Harvard Black Hole Initiative, *Monday Foundations Seminar*, September 2021
Harvard CMSA, *General Relativity Seminar*, September 2021
Sixteenth Marcel Grossman Meeting, *'Radio Astronomy in Space' Session*, July 2021
University of Arizona, **Physics Department Colloquium**, April 2021
Latin American Webinar on Physics, April 2021
Vanderbilt University, *Gravity, Waves and Fluids Initiative Seminar*, April 2021
Princeton University, *High-Energy Theory Seminar*, February 2021

Next-Generation Event Horizon Telescope, *ngEHT Inaugural Science Meeting*, February 2021
Harvard Black Hole Initiative, **2020 Annual Conference**, December 2020
University of Bremen, *Space Science @ Drop Tower Seminar*, October 2020
Harvard Black Hole Initiative, **Colloquium**, September 2020
Princeton University, *Gravity Initiative Seminar*, September 2020
University of California, Davis, *Fields, Strings, Gravity Seminar*, February 2020
Harvard CMSA, *General Relativity Seminar*, February 2020
Institute for Advanced Study, *IAS Astrophysics Seminar*, October 2019
Brown University, *High-Energy Theory Seminar*, September 2018
Harvard Black Hole Initiative, **2018 Annual Conference**, May 2018
McGill University, *High-Energy Theory Seminar*, January 2018
Harvard CMSA, *Mathematical Relativity Workshop*, May 2016
Université Libre de Bruxelles, *Theoretical Physics Seminar*, May 2016
Université Pierre et Marie Curie, Paris VI, *LPTHE Seminar*, January 2016
McGill University, *High-Energy Theory Seminar*, November 2015
Perimeter Institute, *Strong Gravity Seminar*, October 2015

## Top 3 highest-impact publications

⋆ 1) **Universal Interferometric Signatures of a Black Hole's Photon Ring**
M. D. Johnson, A. Lupsasca (**co-first authors**) *et al.* [**362 citations**]
*Science Advances* **6**, no. 12, eaaz1310 (2020), arXiv:1907.04329 [astro-ph.IM]

⋆ 2) **Lensing by Kerr Black Holes**
S. E. Gralla and A. Lupsasca (**alphabetized author list**) [**246 citations**]
*Physical Review D* **101**, no. 4, 044031 (2020), arXiv:1910.12873 [gr-qc]

⋆ 3) **The Shape of the Black Hole Photon Ring: A Precise Test of Strong-Field Gravity**
S. E. Gralla, A. Lupsasca, and D. P. Marrone (**alphabetized author list**) [**224 citations**]
*Physical Review D* **102**, no. 12, 124004 (2020), arXiv:2008.03879 [gr-qc]

## Publications (first-author system)

**Resonances in binary extreme-mass-ratio inspirals: playing the black hole piano**
J. S. Santos, V. Cardoso, A. Lupsasca, J. Natário, and M. van de Meent
*Under review, Accepted for publication in JHEP*, arXiv:2601.02468 [gr-qc]

**Interferometric inference of black hole spin from photon ring size and brightness**
J. Farah, A. Lupsasca, E. Quataert, and M. D. Johnson
*Under review,* arXiv:2509.23628 [astro-ph.HE]

**Black Hole Polarimetry II: The Connection Between Spin and Polarization**
G. N. Wong, A. Chael, A. Lupsasca, and E. Quataert
*The Astrophysical Journal* **997**, no. 1, 113 (2026), arXiv:2509.22639 [astro-ph.HE]

**Why there is no Love in black holes**
A. Lupsasca
*Under review,* arXiv:2506.05298 [gr-qc]

**The Black Hole Explorer: using the photon ring to visualize black hole spacetime**
P. Galison, M. D. Johnson, A. Lupsasca, T. Gravely, and R. Berens
*Proceedings of SPIE* **13902**, 130926R (2024), arXiv:2406.11671 [gr-qc]

**The Black Hole Explorer: photon ring science, detection, and shape measurement**
A. Lupsasca (**first author**) *et al.*
*Proceedings of SPIE* **13902**, 130926Q (2024), arXiv:2406.09498 [gr-qc]

**The Black Hole Explorer: motivation and vision**
M. D. Johnson *et al.*
*Proceedings of SPIE* **13902**, 130922D (2024), arXiv:2406.12917 [astro-ph.IM]

**Photon Ring Interferometric Signatures Beyond The Universal Regime**
H. Jia, E. Quataert, A. Lupsasca, and G. N. Wong
*Physical Review D* **110**, no. 8, 083044 (2024), arXiv:2405.08804 [astro-ph.HE]

**Black Hole Polarimetry I: A Signature of Electromagnetic Energy Extraction**
A. Chael, A. Lupsasca, G. N. Wong, and E. Quataert
*The Astrophysical Journal* **958**, no. 1, 65 (2023), arXiv:2307.06372 [astro-ph.HE]

**Images and photon ring signatures of thick disks around black holes**
F. H. Vincent, S. E. Gralla, A. Lupsasca, and M. Wielgus
*Astronomy & Astrophysics* **667**, A170 (2022), arXiv:2206.12066 [astro-ph.HE]

**Photon ring test of the Kerr hypothesis**
H. Paugnat, A. Lupsasca, F. H. Vincent, and M. Wielgus
*Astronomy & Astrophysics* **669**, A11 (2022), arXiv:2206.02781 [astro-ph.HE]

⋆ **Universal Interferometric Signatures of a Black Hole's Photon Ring**
M. D. Johnson, A. Lupsasca (**co-first authors**) *et al.*
*Science Advances* **6**, no. 12, eaaz1310 (2020), arXiv:1907.04329 [astro-ph.IM]

**Polarization Whorls from M87* at the Event Horizon Telescope**
A. Lupsasca (**first author**), D. Kapec, Y. Shi, D. E. Gates, and A. Strominger
*Proceedings of the Royal Society A* **476**, no. 2237, 20190618 (2020), arXiv:1809.09092 [hep-th]

PUBLICATIONS (ALPHABETIZED AUTHOR LIST)

**Black Hole Vision: An Interactive iOS Application for Visualizing Black Holes**
R. Berens, D. O. Chang, T. Gravely, and A. Lupsasca
*Under review*, arXiv:2603.05810 [gr-qc]

**Single-minus graviton tree amplitudes are nonzero**
A. Guevara, A. Lupsasca, D. Skinner, A. Strominger, and K. Weil on behalf of OpenAI
*Under review*, arXiv:2603.04330 [hep-th]

**Single-minus gluon tree amplitudes are nonzero**
A. Guevara, A. Lupsasca, D. Skinner, A. Strominger, and K. Weil on behalf of OpenAI
*Under review*, arXiv:2602.12176 [hep-th]

**Generalized Couch–Torrence inversions**
P. Charalambous, L. Donnay, and A. Lupsasca
arXiv:2512.18664 [gr-qc]

**Early experiments in accelerating science with GPT-5**
S. Bubeck, C. Coester, R. Eldan, T. Gowers, Y. T. Lee, A. Lupsasca, M. Sawhney, R. J. Scherrer, M. Sellke, B. K. Spears, D. Unutmaz, K. Weil, S. Yin, N. Zhivotovskiy
*OpenAI for Science*, OpenAI

**Gravitational Waves on Kerr Black Holes II:**
**Metric Reconstruction with Cosmological Constant**
R. Berens, T. Gravely, and A. Lupsasca
*Under review*, arXiv:2510.07712 [gr-qc]

**Photon rings in a holographic toy model**
S. Detournay, S. Kanuri, A. Lupsasca, P. Spindel, and Q. Vandermiers
*Journal of High Energy Physics* **2025**, 229 (2025), arXiv:2506.07989 [hep.th]

**Explanation for the absence of secondary peaks in black hole light curve autocorrelations**
A. Cárdenas-Avendaño, C. Gammie, and A. Lupsasca
*Physical Review Letters* **133**, no. 13, 131402 (2024), arXiv:2406.04176 [astro-ph.HE]

**Assessing the impact of instrument noise and astrophysical fluctuations on measurements of the first black hole photon ring**
A. Cárdenas-Avendaño, L. Keeble, and A. Lupsasca
*Physical Review D* **109**, no. 12, 124052 (2024), arXiv:2404.01083 [gr-qc]

**Gravitational Waves on Kerr Black Holes I: Reconstruction of Linearized Metric Perturbations**
R. Berens, T. Gravely, and A. Lupsasca
*Classical and Quantum Gravity* **41**, no. 19, 195004 (2024), arXiv:2403.20311 [gr-qc]

**A Beginner's Guide to Black Hole Imaging and Associated Tests of General Relativity**
A. Lupsasca, D. R. Mayerson, B. Ripperda, and S. Staelens
*Recent Progress on Gravity Tests* (Chapter **6**), Springer, Singapore (2024), arXiv:2402.01290 [gr-qc]

**Black hole bulk-cone singularities**
M. Dodelson, C. Iossa, R. Karlsson, A. Lupsasca, and A. Zhiboedov
*Journal of High Energy Physics* **2024**, 46 (2024), arXiv:2310.15236 [hep-th]

**Prediction for the interferometric shape of the first black hole photon ring**
A. Cárdenas-Avendaño and A. Lupsasca
*Physical Review D* **108**, no. 4, 064043 (2023), arXiv:2305.12956 [gr-qc]

**Adaptive Analytical Ray Tracing of Black Hole Photon Rings**
A. Cárdenas-Avendaño, A. Lupsasca, and H. Zhu
*Physical Review D* **107**, no. 4, 043030 (2023), arXiv:2211.07469 [gr-qc]

**Photon Rings Around Warped Black Holes**
D. Kapec, A. Lupsasca, and A. Strominger
*Classical and Quantum Gravity* **40**, no. 9, 095006 (2023), arXiv:2211.01674 [gr-qc]

**Holography of the Photon Ring**
S. Hadar, D. Kapec, A. Lupsasca, and A. Strominger
*Classical and Quantum Gravity* **39**, no. 21, 215001 (2022), arXiv:2205.05064 [gr-qc]

**Observing the Inner Shadow of a Black Hole: A Direct View of the Event Horizon**
A. Chael, M. D. Johnson, and A. Lupsasca
*The Astrophysical Journal* **918**, no. 1, 6 (2021), arXiv:2106.00683 [astro-ph.HE]

**Extreme Black Hole Anabasis**
S. Hadar, A. Lupsasca, and A. Porfyriadis
*Journal of High Energy Physics* **2021**, 223 (2021), arXiv:2012.06562 [hep-th]

**Photon Emission from Circular Equatorial Kerr Orbiters**
D. E. A. Gates, S. Hadar, and A. Lupsasca
*Physical Review D* **103**, no. 4, 044050 (2021), arXiv:2010.07330 [gr-qc]

**Photon Ring Autocorrelations**
S. Hadar, M. D. Johnson, A. Lupsasca, and G. N. Wong
*Physical Review D* **103**, no. 10, 104038 (2021), arXiv:2010.03683 [gr-qc]

**Maximum Observable Blueshift from Circular Equatorial Kerr Orbiters**
D. E. A. Gates, S. Hadar, and A. Lupsasca
*Physical Review D* **102**, no. 10, 104041 (2020), arXiv:2009.03310 [gr-qc]

⋆ **The Shape of the Black Hole Photon Ring: A Precise Test of Strong-Field Gravity**
S. E. Gralla, A. Lupsasca, and D. P. Marrone
*Physical Review D* **102**, no. 12, 124004 (2020), arXiv:2008.03879 [gr-qc]

**On the Observable Shape of Black Hole Photon Rings**
S. E. Gralla and A. Lupsasca
*Physical Review D* **102**, no. 12, 124003 (2020), arXiv:2007.10336 [gr-qc]

**Universal Polarimetric Signatures of the Black Hole Photon Ring**
E. Himwich, M. D. Johnson, A. Lupsasca, and A. Strominger
*Physical Review D* **101**, no. 8, 084020 (2020), arXiv:2001.08750 [gr-qc]

**Null Geodesics of the Kerr Exterior**
S. E. Gralla and A. Lupsasca
*Physical Review D* **101**, no. 4, 044032 (2020), arXiv:1910.12881 [gr-qc]

⋆ **Lensing by Kerr Black Holes**
S. E. Gralla and A. Lupsasca
*Physical Review D* **101**, no. 4, 044031 (2020), arXiv:1910.12873 [gr-qc]

**Particle Motion Near High-Spin Black Holes**
D. Kapec and A. Lupsasca
*Classical and Quantum Gravity* **37**, no. 1, 015006 (2019), arXiv:1905.11406 [hep-th]

**Critical Emission from a High-Spin Black Hole**
A. Lupsasca, A. P. Porfyriadis, and Y. Shi
*Physical Review D* **97**, no. 6, 064017 (2018), arXiv:1712.10182 [gr-qc]

**Observational Signature of High Spin at the Event Horizon Telescope**
S. E. Gralla, A. Lupsasca, and A. Strominger
*MNRAS* **475**, no. 3, 3829–3853 (2018), arXiv:1710.11112 [astro-ph.HE]

**Inclined Pulsar Magnetospheres in General Relativity: Polar Caps for the Dipole, Quadrudipole and Beyond**
S. E. Gralla, A. Lupsasca, and A. Philippov
*The Astrophysical Journal* **851**, no. 2, 137 (2017), arXiv:1704.05062 [astro-ph.HE]

**Force-Free Foliations**
G. Compère, S. E. Gralla, and A. Lupsasca
*Physical Review D* **94**, no. 12, 124012 (2016), arXiv:1606.06727 [math-ph]

**Pulsar Magnetospheres: Beyond the Flat Spacetime Dipole**
S. E. Gralla, A. Lupsasca, and A. Philippov
*The Astrophysical Journal* **833**, no. 2, 258 (2016), arXiv:1604.04625 [astro-ph.HE]

**Near-horizon Kerr Magnetosphere**
S. E. Gralla, A. Lupsasca, and A. Strominger
*Physical Review D* **93**, no. 10, 104041 (2016), arXiv:1602.01833 [hep-th]

**Electromagnetic Jets from Stars and Black Holes**
S. E. Gralla, A. Lupsasca, and M. J. Rodriguez
*Physical Review D* **93**, no. 4, 044038 (2016), arXiv:1504.02113 [gr-qc]

**Note on Bunching of Field Lines in Black Hole Magnetospheres**
S. E. Gralla, A. Lupsasca, and M. J. Rodriguez
*Physical Review D* **92**, no. 4, 044053 (2015), arXiv:1504.02112 [gr-qc]

**Exact Solutions for Extreme Black Hole Magnetospheres**
A. Lupsasca and M. J. Rodriguez
*Journal of High Energy Physics* **2015**, 90 (2015), arXiv:1412.4124 [hep-th]

**Force-Free Electrodynamics around Extreme Kerr Black Holes**
A. Lupsasca, M. J. Rodriguez, and A. Strominger
*Journal of High Energy Physics* **2014**, 185 (2014), arXiv:1406.4133 [hep-th]

**Quasinormal Quantization in de Sitter spacetime**
D. L. Jafferis, A. Lupsasca, V. Lysov, G. S. Ng, and A. Strominger
*Journal of High Energy Physics* **2015**, 4 (2015), arXiv:1305.5523 [hep-th]